

Customer Clustering & Purchasing Behavior

Business Analytics Project

Data Source: GOA Supermarket customer dataset — 100 customer records

This project analyzes customer data from GOA Supermarket to identify purchasing-behavior segments and provide data-driven recommendations for targeted marketing and customer retention.

Dataset Provenance

The dataset used in this project is presented as a GOA Supermarket dataset containing 100 customer records. It includes customer age, annual income, purchase frequency, average purchase value, and category preference. These records form the underlying business data analyzed throughout the project.

The analysis was performed for GOA Supermarket to understand differences between customer groups and translate those findings into practical marketing and customer-retention strategies.

Business Objective

GOA Supermarket needed a clearer understanding of how its customers differ in purchasing behavior so that marketing and customer-engagement decisions could move beyond a one-size-fits-all approach.

Objectives



Identify distinct customer groups within GOA Supermarket's customer dataset.



Understand differences in demographic and purchasing characteristics across the groups.



Provide a data-driven basis for differentiated marketing and customer engagement.



Support customer-retention planning by identifying different customer types and purchasing patterns.

Analytical Approach

The workflow included data cleaning, outlier review, standardization of numerical variables, categorical encoding, DBSCAN and Agglomerative (hierarchical) Clustering, elbow-method analysis, segment-level analysis, and visualization.

Results

The implemented Agglomerative Clustering model produced three customer segments.

Segment-level analysis examined differences in income, purchase frequency, average purchase

value, age, and category preference.

The resulting segments provide GOA Supermarket with a structured view of customer differences that can be used to design differentiated promotions, engagement approaches, and retention initiatives.

Business Impact



Replaces a one-size-fits-all customer view with three data-driven customer segments.



Provides a basis for more targeted promotional strategies.



Supports customer-engagement and retention planning based on observed customer characteristics.



Gives management a clearer analytical framework for understanding customer purchasing behavior.

Tools Used

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

Project outcome: A customer-segmentation analysis based on GOA Supermarket's original 100-customer dataset, providing a data-driven foundation for differentiated marketing and customer-retention decisions.